The Ultrasonic Anemometer: A deeper than usual view

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BY NC ND

Abstract

In commercials and popular science the operation of pulse-type three dimensional ultrasonic anemometer is presented in a highly simplified manner, namely assuming airflow is parallel to the measurement axis, and sound pulses can be located exactly (that is with no uncertainty) in time.

Descriptions of how actually a specific model of ultrasonic anemometer work do exist, in the form of design documents by sensor manufacturers. These report, covered by industrial secret, cannot be accessed by random readers (and the scientific community).

This note attempts to fill the gap, providing a level of information beyond commercials, allowing scientists and advanced users to better understand the nature of the numbers they are using – without infringing patents or industrial secrets.

Part of this knowledge can be derived by anyone by applying simple reasoning. Another part can be obtained by measurement, using commercial off-the-shelf technology. Both sources can be considered public.

1.The simplified presentation

Figure 1 - An example of 3D ultrasonic anemometer: the uSonic-3 Multipath, by Metek GmbH

Ultrasonic anemometers, especially in their three-dimensional incarnation, are gaining momentum in a variety of applications ranging from radio-propagation studies to civil engineering passing through air quality – and many more.

But also two-dimensional models, often marketed as low-maintenance hassle-free replacements of mechanical cup-vane or propeller-vane anemometers, are now widely diffused: they are based on the same physical principle of their 3D counterparts, and differ from them in using only two measurement axes instead of three, and a firmware designed to simulate the behavior of mechanical sensors (except vulnerability to wind calms, situations in which wind speed is lower than the anemometers' *activation threshold*).

In the "usual" simplified presentation of ultrasonic anemometer a single measurement axis is considered, as shown in Figure 2.

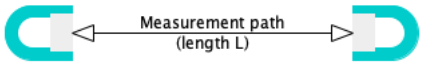

Figure 2 - Ultrasonic anemometer measurement path

The path, of a precisely known length L, is delimited by two heads capable to operate as both receiver or transmitter; during measurements they alternate between the two roles so that when one transmits, the other receives.

Transmissions and receptions regard, then, "acoustic pulses". We imagine these pulses as high frequency, as of the sensor name, "ultrasonic anemometer". But this is really irrelevant, at this description level: pulses can (*should*) be imagined infinitely focused, and detectable in zero time.

The latter requirement is critical: all reasoning bases on the possibility to measure as precisely as wanted the time between pulse emission and pulse reception.

Now. Let's label the leftmost head in Figure 1 as "A", and "B" the rightmost. We than can say of the two times t_{AB} and t_{BA} the two flight times assuming respectively A is transmitter and B receiver, or B transmitter and A receiver. By the assumptions we made, we can say they both are *exact*.

In still air, of course it is $t_{AB} = t_{BA} = \frac{L}{c}$ where c is the speed of sound.

In Nature, air is never still. And so, we can imagine some non-zero wind is present. By necessity of the simplified presentation, we have to imagine this non-zero wind is directed along the measurement path, from A (operating as transmitter) to B (receiver). This situation is shown in Figure 3.

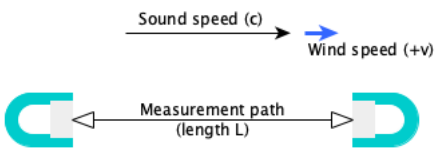

Figure 3 - There is wind, blowing from transmitter to receiver

Then we imagine A and B exchange their roles, while the wind remains unchanged: that is, this time B is the transmitter, A the receiver, and wind again goes from A to B. This is exactly the same as saying the pulse is still released by transmitter A, with wind proceeding along the *opposite* direction.

Like we can see in Figure 4.

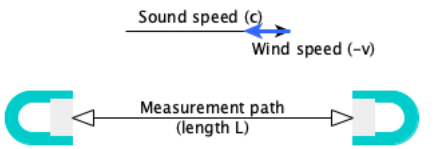

Figure 4 - Wind is still present, but from receiver to transmitter

Can we say something on pulses flight time? Of course we do. In the first case wind and sound speed add, so that

$$t_{A\to B} = \frac{L}{c+v}$$

In the second,

$$t_{B\to A} = \frac{L}{c-v}$$

Let us now take

$$\frac{1}{t_{A\to B}} - \frac{1}{t_{B\to A}} = \frac{c+v}{L} - \frac{c-v}{L} = 2\frac{v}{L}$$

that is,

$$v = \frac{L}{2}\left(\frac{1}{t_{A\to B}} - \frac{1}{t_{B\to A}}\right)$$

In the same way,

$$\frac{1}{t_{A\to B}} + \frac{1}{t_{B\to A}} = \frac{c+v}{L} + \frac{c-v}{L} = 2\frac{c}{L}$$

or

$$c = \frac{L}{2}\left(\frac{1}{t_{A\to B}} + \frac{1}{t_{B\to A}}\right)$$

The previous relations allow to express the wind velocity and sound speed by a proper combination of a constant $(L/2)$ and the sum or difference of the reciprocal propagation times.

And this is the way ultrasonic anemometer theory is typically presented.

2.A more realistic presentation

2.1.Issues with the simplified presentation

All the preceding equations have been derived under severely restrictive assumptions. The one possibly more invasive regards the direction of wind, parallel to measurement axis.

A three-dimensional ultrasonic anemometer has three measurement paths, arranged to form what mathematicians name a *basis* of the three-dimensional space $\mathbb{R}^3$ – in practice, they do not lie on a same plane, or even a same straight direction. But so, would any one of those measurement paths be parallel to wind the other two could necessarily not.

Said differently, in a real-world ultrasonic anemometer the wind will have both a parallel and a *transversal* component, respect to measurement axes. What happens due to this transversal component? Do the sonic equations remain the same?

Fortunately, the problem of conceptualizing this eventuality is not that hard. First, we observe the sound pulse does not propagate along a straight line, but rather on a spherical wavefront initially centered on emission point.

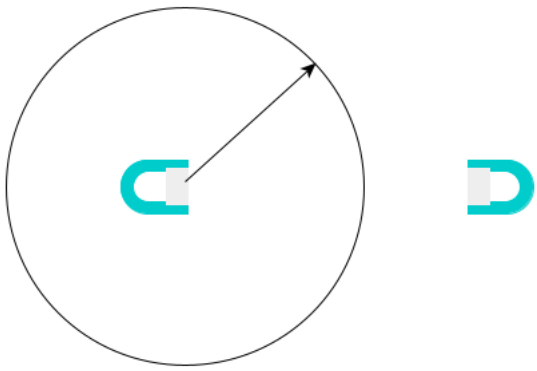

Figure 5 - Sound pulse propagation in absence of wind; the arrow indicates "one" of the infinite propagation directions

In Figure 5 we can see the wavefront at some specific time in still air, assuming the pulse is emitted by head A.

Figure 6 shows instead what happens in case wind is present. After emission, the entire wavefront is advected adrift, and if a wind component exists along the measurement path its arrival at head B will occur sooner or later than expected would propagation occur only after sound speed c.

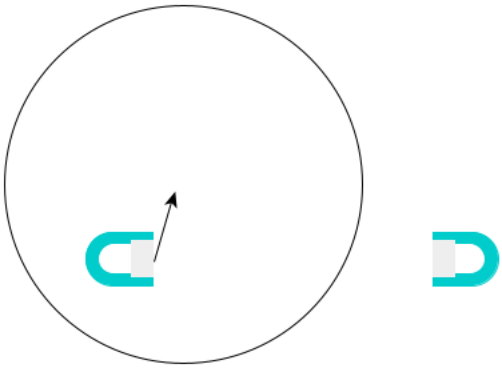

Figure 6 - Wavefront propagation in presence of wind

As figures 5 and 6 suggest, the wavefront will not hit the receiving head surface as a plane, and this effect will be more and more pronounced as the transversal wind component gets higher. Both these effects are negligible however, at least within the measurement range of ultrasonic anemometers, in the order of 60 m/s – a small fraction of the more than 300 m/s of c.

Hitting the receiver' surface at an angle however *may* be an issue, as energy is far from being distributed evenly over the wave surface. Intended as loudspeakers, emitting heads are strongly directional. Directionality is further enhanced by the high frequency of sound pulses. Now, in the advected wavefront the highest power occurs in the direction of the measurement path as it was on emission instant. The hitting part of the wavefront, so, will be off axis – and more so in case of string transversal wind. The ensuing lower power implies an attenuated receiver answer, and then a lower signal-to-noise.

All this given, in addition to the geometrical question, the presence of a transversal wind component should be addressed in some way.

2.2. The geometrical description of what happens

Sooner or later, if airflow is not "supersonic and opposite to the sound pulse receiver", the wavefront will reach the receiver, which will react by generating an electrical signal triggering the conclusion of the time measurement interval.

But in the same time, the wavefront will have advected as shown in Figure 7.

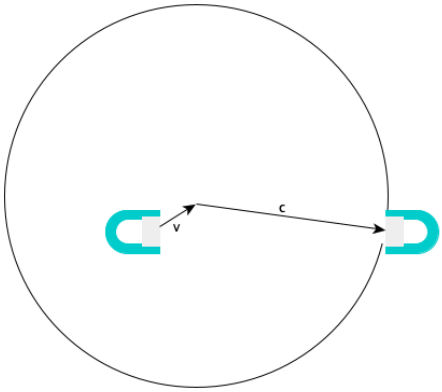

Figure 7 - Wavefront will reach receiver after some time, during which it will have advected due to wind

The situation depicted in Figure 7 is universal, that is, independent on the wind vector provided the latter is not supersonic.

2.3.Analytic version

To express the geometric version in formulae we first introduce a reference system in which the x axis is directed along the measurement axis, from head A to B. We then fix the y axis perpendicular to the x, and such that the wind vector $\mathbf{v}$ can be expressed as

$$\mathbf{v} = [v_x, v_y, 0]$$

(we may enforce this condition by rotating the reference frame around the x axis until the z component of velocity is 0, whatever its original value).

It is now useful to introduce the *deviation angle* θ through the condition

$$\sin\theta = \frac{v_y}{|\mathbf{v}|}$$

where I've indicated as usual with $|\mathbf{v}|$ the vector length $\sqrt{v_x^2 + v_y^2 + v_z^2} = \sqrt{v_x^2 + v_y^2 + 0} = \sqrt{v_x^2 + v_y^2}$. With such position we may write

$$v_y = |\mathbf{v}| \sin\theta$$

(not a great discovery, just a rewrite of the previous formula) and

$$v_x = |\mathbf{v}| \cos\theta$$

Now let us consider the wavefront. We have already seen in the preceding section that it is "spherical" (it isn't, due to the finite extension of the sound emitting surface of heads, but the difference is small, and we can safely neglect it), with its center translating adrift the wind. As we have already decided to choose our reference frame in a convenient way we do the final step and assume its origin at center of the emission surface of head A. Imagining the sound pulse emission occurs at $t = 0$ we get for the wavefront center the component expression

$$W = [v_x t, v_y t, 0]$$

The wavefront expands at velocity c, the speed of sound, while it translates. Its section on the xy plane if then the circumference having equation

$$[v_x t + ct\cos\alpha\,, v_y t + ct\sin\alpha\,, 0]$$

with α an angle assuming values $0 \le \alpha < 2\pi$.

We now need to identify the hitting condition, which is not that difficult if we notice that in the reference frame we have so carefully selected the center of head B's receiving surface has coordinates

$$[0, L, 0]$$

By equating the last two we get the hitting condition:

$$\begin{cases} v_x t + ct\cos\alpha = L \\ v_y t + ct\sin\alpha = 0 \end{cases}$$

This is an equation in the two variables t (very interesting) and α (worthless), with two solutions[1]

$$t_1 = \frac{-L\left(v_x + \sqrt{c^2 - v_y^2}\right)}{c^2 - |\mathbf{v}|^2} \qquad \alpha_1 = \tan^{-1}\frac{v_y}{\sqrt{c^2 - v_y^2}} + 2k\pi$$

and

$$t_2 = \frac{-L\left(v_x - \sqrt{c^2 - v_y^2}\right)}{c^2 - |\mathbf{v}|^2} \qquad \alpha_2 = \tan^{-1}\frac{-v_y}{\sqrt{c^2 - v_y^2}} + 2k\pi$$

where $k = \cdots, -3, -2, -1, 0, 1, 2, 3, \cdots$ is an arbitrary integer. As mentioned above, the values α_1, α_2 have no practical interest. Not so for t_1 and t_2, one of these is the one we're interested in, the one positive, that is t_2.

The same reasoning may be repeated if emission takes place from B, A acting as receiver. But in this case the equations become

$$\begin{cases} v_x t + ct\cos\alpha = -L \\ v_y t + ct\sin\alpha = 0 \end{cases}$$

with solutions in t

$$t_3 = \frac{L\left(v_x + \sqrt{c^2 - v_y^2}\right)}{c^2 - |\mathbf{v}|^2}$$

and

$$t_4 = \frac{L\left(v_x - \sqrt{c^2 - v_y^2}\right)}{c^2 - |\mathbf{v}|^2}$$

This time the positive solution is t_3 and we may conclude

$$t_{AB} = t_2$$

and

$$t_{BA} = t_3$$

Let us now consider

$$\frac{1}{t_{AB}} - \frac{1}{t_{BA}} = \frac{c^2 - |\mathbf{v}|^2}{-L\left(v_x - \sqrt{c^2 - v_y^2}\right)} - \frac{c^2 - |\mathbf{v}|^2}{L\left(v_x + \sqrt{c^2 - v_y^2}\right)}$$

We may rewrite the right member as

[1] Found and confirmed with Wolfram Research's *Mathematica*. ☺

$$(c^2 - |\mathbf{v}|^2)\left[\frac{1}{-L\left(v_x - \sqrt{c^2 - v_y^2}\right)} - \frac{1}{L\left(v_x + \sqrt{c^2 - v_y^2}\right)}\right] =$$

$$= (c^2 - |\mathbf{v}|^2)\left[\frac{-L\left(v_x + \sqrt{c^2 - v_y^2}\right) - L\left(v_x - \sqrt{c^2 - v_y^2}\right)}{L\left(v_x - \sqrt{c^2 - v_y^2}\right)L\left(v_x + \sqrt{c^2 - v_y^2}\right)}\right] =$$

$$= (c^2 - |\mathbf{v}|^2)\left[\frac{-Lv_x - L\sqrt{c^2 - v_y^2} - Lv_x + L\sqrt{c^2 - v_y^2}}{L\left(v_x - \sqrt{c^2 - v_y^2}\right)L\left(v_x + \sqrt{c^2 - v_y^2}\right)}\right] =$$

$$= (c^2 - |\mathbf{v}|^2)\left[\frac{-2Lv_x}{L\left(v_x - \sqrt{c^2 - v_y^2}\right)L\left(v_x + \sqrt{c^2 - v_y^2}\right)}\right] =$$

$$= (c^2 - |\mathbf{v}|^2)\left[\frac{-2v_x}{L\left(v_x - \sqrt{c^2 - v_y^2}\right)\left(v_x + \sqrt{c^2 - v_y^2}\right)}\right] =$$

$$= (c^2 - |\mathbf{v}|^2)\left[\frac{-2v_x}{L\left(v_x^2 - c^2 + v_y^2\right)}\right] =$$

$$= (c^2 - |\mathbf{v}|^2)\left[\frac{-2v_x}{L(|\mathbf{v}|^2 - c^2)}\right] =$$

$$= \frac{2v_x}{L}$$

This is very interesting: if the wind component v_y transversal to the measurement path is smaller than c, which is guaranteed for subsonic airflow, the difference between the reciprocals of the propagation times is proportional to the wind component along the measurement path, and independent of the transversal component.

This is the same we saw in the simplified presentation.

Now, let's take the sum of reciprocals:

$$\frac{1}{t_{AB}} + \frac{1}{t_{BA}} = \frac{c^2 - |\mathbf{v}|^2}{-L\left(v_x - \sqrt{c^2 - v_y^2}\right)} + \frac{c^2 - |\mathbf{v}|^2}{L\left(v_x + \sqrt{c^2 - v_y^2}\right)}$$

This too can be rewritten:

$$\frac{c^2 - |\mathbf{v}|^2}{-L\left(v_x - \sqrt{c^2 - v_y^2}\right)} + \frac{c^2 - |\mathbf{v}|^2}{L\left(v_x + \sqrt{c^2 - v_y^2}\right)} =$$

$$= (c^2 - |\mathbf{v}|^2)\frac{1}{-L\left(v_x - \sqrt{c^2 - v_y^2}\right)} + \frac{1}{L\left(v_x + \sqrt{c^2 - v_y^2}\right)} =$$

$$= (c^2 - |\mathbf{v}|^2)\left[\frac{-L\left(v_x + \sqrt{c^2 - v_y^2}\right) + L\left(v_x - \sqrt{c^2 - v_y^2}\right)}{L\left(v_x - \sqrt{c^2 - v_y^2}\right)L\left(v_x + \sqrt{c^2 - v_y^2}\right)}\right] =$$

$$= (c^2 - |\mathbf{v}|^2)\left[\frac{-\left(v_x + \sqrt{c^2 - v_y^2}\right) + \left(v_x - \sqrt{c^2 - v_y^2}\right)}{L\left(v_x - \sqrt{c^2 - v_y^2}\right)\left(v_x + \sqrt{c^2 - v_y^2}\right)}\right] =$$

$$= (c^2 - |\mathbf{v}|^2)\left[\frac{-v_x - \sqrt{c^2 - v_y^2} + v_x - \sqrt{c^2 - v_y^2}}{-L(c^2 - |\mathbf{v}|^2)}\right] =$$

$$= \frac{2\sqrt{c^2 - v_y^2}}{L}$$

This is *not* the same we've seen in the simplified presentation. It may come close, if $|v_y| \ll c$, but some difference is.

In this all, it should bear in mind we assumed emission and reception to occur *pointwise* and *without any time uncertainty*. My presentation is also simplified, just by a lesser extent.

3.Finite head size and its consequences

As mentioned in sections 2.1 and 2.3, the heads of any ultrasonic anemometer are not pointwise, and their finite size should be accounted for.

Instead of addressing this issue analytically, a measurement approach was used by means of the author's recording rig, composed by a RØDE STG3 directional microphone and a Zoom F3 digital recorder. Although rated for audible range, the sensitivity of STG3 extends to the near ultrasound; the Zoom F3 has been programmed with a sampling rate of 192 kHz and 32 bit floating point samples.

The microphone was pointed to one of the heads along one of the three measurement directions of a Metek USA-1, close nearby to the head on opposite end. The anemometer was programmed with a sampling rate of 20 Hz, and 1-out-of-2 packets averaging.

The test was conducted by the author in her laboratory, with normal heating

The time plot of one of the samples is given in Figure 8.

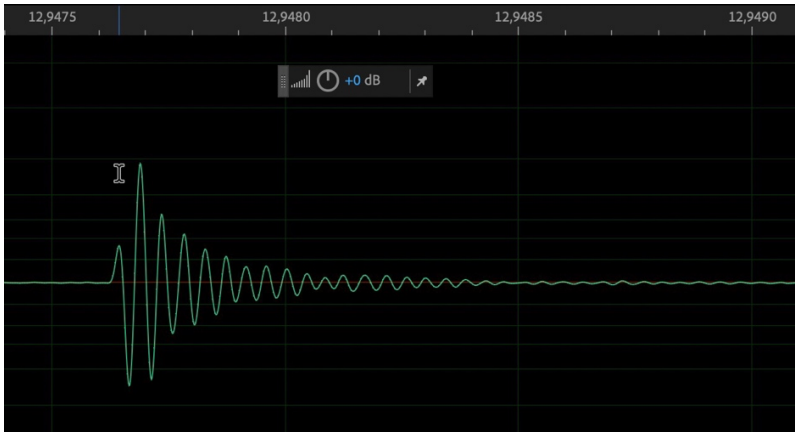

Figure 8 - Sound pulse from an USA-1 ultrasonic anemometer

In Figure 8 we can distinguish three phases:

1. An *attack* phase, about one full period long, during which the sound pressure reaches its maximum level starting from 0.
2. A *sustain* phase, half period long, in which the pulse sound pressure is at maximum.
3. A *decay* phase, in which the pulse decreases its amplitude exponentially.

The terms "attack", "sustain" and "decay" are analogous to the same terms used in electronic music, but with a different meaning, or better time scale. In the music case the three phases have a directly perceptible duration. This is not the case with ultrasonic anemometer pulses, whose durations are not perceptible – at least at a conscious level.

Figure 8 shows an “ideal” situation in still air. The presence of transversal wind causes some attenuation, but the overall shape of a pulse in the time domain remains unchanged.

It should also be noted the signal in Figure 8 presents markers whose position in time is easy to find – namely zero crossings. The passages through zero from number two to five could be nice target for an automatic clock.

This, coupled with a careful hardware and firmware design, allows to prevent the direst consequences of the issues mentioned in the previous sections.

4.Conclusions

Description stops here, at the fringe of industrial secrets and patents. It can in no way considered “exhaustive”, but the hope is it is sufficient to understand the process of ultrasonic wind measurement at a somewhat deeper level than usual, along with an awareness that ultrasonic anemometers are indeed “deep” sensors, whose construction cannot be improvised.

Of course, there is much more to say (some of which the author is not authorized to report).

But would you need to understand somewhat more about the sensor application you may contact the author at her e-mail address, patti.favaron@gmail.com